Decent Coaching Classes

X Semi

Academic Conquest - VII

Subject: Science and Technology (Ch. 1 Gravitation) **Marks: 30**

Note: Figures to the right indicate full marks. Write neat and labelled diagrams wherever necessary.

Q.1 Choose the correct alternative and rewrite the sentence. **[5]**

1. The SI unit of weight is _____.
   (a) kg (b) Newton (c) m/s^2 (d) Pascal
2. The value of acceleration due to gravity (g) on the surface of the earth is about _____.
   (a) 0.98 m/s^2 (b) 9.8 m/s^2 (c) 98 m/s^2 (d) 6.67 m/s^2
3. According to Kepler's first law, the orbit of a planet is _____.
   (a) a circle (b) a straight line (c) an ellipse (d) a square
4. The law of universal gravitation was given by _____.
   (a) Kepler (b) Galileo (c) Newton (d) Archimedes
5. The escape velocity on the surface of the earth is about _____.
   (a) 11.2 km/s (b) 2.37 km/s (c) 9.8 km/s (d) 100 km/s

Q.2 Answer in one sentence. (Any 3 of 4) **[3]**

(a) State Kepler's second law.
(b) What is free fall?
(c) What is centripetal force?
(d) Who first measured the value of the universal gravitational constant (G) experimentally?

Q.3 Give scientific reasons. (Any 2 of 3) **[4]**

(a) The weight of an object on the moon is less than its weight on the earth.
(b) True free fall is possible only in vacuum.
(c) A stone thrown vertically upwards falls back to the ground.

Q.4 Answer the following questions in brief. (Any 3 of 4) **[6]**

(a) State Kepler's first law and third law.
(b) Distinguish between mass and weight (two points).
(c) Write the formula for the acceleration due to gravity (g) in terms of the mass and radius of the earth, and state its value on the surface of the earth.
(d) What is escape velocity? Write its value for the earth.

Q.5 Answer the following questions. (Any 2 of 3) **[6]**

(a) Explain centripetal force with the help of the activity of a stone tied to a string and whirled in a circle.
(b) State Newton's universal law of gravitation and write its mathematical formula.
(c) Explain how the value of g changes as we go above the surface of the earth.

Q.6 Solve the following numerical examples. (Any 2 of 3) **[6]**

(a) The mass of a body is 60 kg. Calculate its weight on the earth and on the moon. (g on the earth = 10 m/s^2, g on the moon = 1.6 m/s^2)
(b) A stone is dropped from rest and takes 3 s to reach the ground. Calculate the velocity of the stone on reaching the ground and the height from which it was dropped. (g = 10 m/s^2)
(c) Calculate the gravitational force between two bodies of masses 10 kg and 20 kg kept 1 m apart. ($G = 6.67 \times 10^{-11}$ N m^2 kg^{-2})